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ORIGINAL PAPER

A novel multiple moving objects recognition and segmentation based on dense optical flow and K-means clustering

Jing Ding[1,2] · Zhen Zhang[1] · Yangyang Wang[3]

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Abstract
Moving objects detection has always been a prominent research topic for the computer vision and image processing field. In recent years, many methods of moving objects detection in optical imaging have been proposed. However, the high-precision detection and recognition of multiple moving objects is still quite challenging. Many key problems that restrict the performance of multiple moving objects detection, such as dynamic background, complex environment, different objects classification, and detection accuracy, have not been effectively solved. This study proposes a dense optical flow multiple moving objects recognition and extraction method to overcome these problems. Unlike most moving objects detection methods based on sparse optical flow, we propose a multiple moving objects detection and recognition method which combines an optical flow histogram and K-means clustering analysis based on dense optical flow. First, we utilize an enhanced optical flow calculation and feature threshold processing to segment the moving objects foreground region. Subsequently, according to the histogram of optical flow amplitude and directions, K-means iterative clustering is utilized to cluster the foreground optical flow. Different moving objects can be recognized and segmented using different optical flow amplitudes and direction. The implementation results indicate that the proposed method exhibits more optimal results for both rigid and non-rigid multiple targets with small internal optical flow difference in static and dynamic environments.

Keywords Multiple moving objects · Objects detection · Clustering analysis · K-means · HS optical flow

1 Introduction

Moving objects recognition and segmentation has always been a crucial research field in computer image vision [1–4]. The technology identifies and extracts moving objects from complex background environments in image sequences or videos [5–7], and it can be utilized in many fields such as intelligent monitoring [8], intelligent transportation [9], unmanned driving [10], and target tracking [11]. According to the number of objects detection, moving object detection can be divided into single object detection and multi-objects detection. Currently, the single moving objects detection technology is relatively mature. The research on multiple moving objects detection mainly focuses on the preliminary extraction of multiple moving objects, and there is no further classification between different moving objects. There are few studies [12–14] on the accurate extraction of multiple moving objects and the recognition of different individuals. Multiple moving objects detection is not only subjected to problems such as dynamic background, illumination change, complex background, and image 'ghost'

Zhen Zhang and Yangyang Wang contributed equally to this work.

✉ Jing Ding
austdj@aust.edu.cn

Zhen Zhang
zhangzhen@aust.edu.cn

Yangyang Wang
454260183@qq.com

1 School of Geomatics, Anhui University of Science and Technology, Huainan 232001, China

2 Coal Industry Engineering Research Center of Mining Area Environmental And Disaster Cooperative Monitoring, Anhui University of Science and Technology, Huainan 232001, China

3 Shandong Provincial No. 4 Institute of Geological and Mineral Survey, Weifang 261021, China

[15], but also needs to solve the problem associated with the recognition and classification of different objects. The high-precision detection and recognition of multiple moving objects in complex environments remains challenging.

Based on different detection techniques, moving objects detection can be divided into three main methods: consecutive frame difference [16], background subtraction [17], and optical flow [18]. The consecutive frame difference method utilizes image difference to detect moving objects, which is not applicable when the scene is moving, and the error of detection results is large. Background subtraction utilizes the background model and the image difference to detect the moving objects. The background model should be estimated first; furthermore, the calculation is more complicated, and it is disturbed by the image 'ghost ' problem. The optical flow method extracts image motion by estimating the image optical flow vector field, which exhibits more optimal detection performance when subjected to scale or morphological changes, and can also generate motion parameters, which is more conducive to the classification and recognition of multiple moving objects.

The objects detection based on the optical flow method includes Horn–Schunck (HS) optical flow [19] and Lucas–Kanade (LK) optical flow [20]. The object detection based on the LK sparse optical flow field adopts the sparse points optical flow feature clustering; although multiple objects detection is possible, the results are mostly sparse points, and the accuracy is low. However, most of the existing moving object detection and extraction algorithms based on the HS dense optical flow method only initially detect and segment the foreground area of all moving objects area jointly; thus, different moving objects are not further refined and segmented. Consequently, multiple objects will characterize the foreground area, and the single object cannot be recognized. Based on the research on the optical flow and robust feature points thresholding processing moving objects detection [21], this study further analyses the detailed recognition and segmentation of multiple moving objects. According to the large difference in the optical flow amplitude and direction between different moving objects, we propose a multiple moving objects recognition and extraction algorithm based on an optical flow histogram and K-means clustering, which can be utilized for multiple moving objects detection and the recognition of dense image optical flow. The main contributions of this study are as follows:

- This study represents an optical flow clustering method for a dense optical flow field.
- This method realizes the detailed recognition and segmentation extraction of different moving objects.
- This method is suitable for a variety of common dense optical flow estimation methods. Even if the optical flow estimation results are unreliable, it can also assist in the detection and recognition of multiple moving objects and enhance the detection efficiency.

2 Related work

The moving objects detection based on optical flow is aimed at segmenting the moving targets from the complex environments using the estimated optical flow information [22, 23]. By classifying and summarizing a large number of optical flow moving objects detection algorithms, moving objects detection can be divided into three methods: objects detection based on threshold processing, objects detection based on optical flow clustering, and a combination that entails other methods.

The moving objects detection based on optical flow threshold processing is aimed at transforming the flow field into grayscale information, thresholding the grayscale image, and converting it into the binary image of moving objects. The most crucial procedure that characterizes optical flow thresholding is the determination of the optical flow threshold, which has been analysed by many scholars in recent years. Qi et al. [24] extracted foreground targets by artificially selecting and setting optical flow thresholds, which could not address the batch processing requirements of a large number of images and real-time performance of the algorithm. Otsu always utilized an adaptive segmentation algorithm to extract the binary images of moving objects [25–28]. The extraction result is satisfactory under the static background, but poor under the moving background environment. Sengar et al. [29, 30] calculated the optical flow between three consecutive images; subsequently, the scholars utilized Otsu's method to extract the area. This method exhibits considerable accuracy; however, the optical flow repeat calculation immensely decrease the efficiency. Han et al. [31] determined the threshold using a linear and nonlinear mixed function combining the optical flow feature vector and the optical flow scalar value. Huang et al. [32]. utilized the Homography Matrix with the RANSAC method to calculate and optimize the environment model, which can automatically calculate the threshold, and the model to automatically adjust as the motion state changes. The current methods can extract the moving targets quickly; however, they should respectively analyse the background static or motion, and for the scenario that entails multiple moving objects, they can segment only the foreground area of all moving objects; different moving objects cannot be recognized.

The moving objects detection based on optical flow clustering [33–35] obtains the moving targets foreground area by clustering similar optical flow vectors. The gray image and the distance of estimated HS dense optical flow vectors are utilized as the spatial and temporal information of the image [36]. The image is divided into four initial feature regions, and the corresponding initial clustering centre vector is set for K-means iterative clustering, whose results contain noises. Compared with the threshold segmentation method, the optical flow clustering algorithm can recognize moving objects in a moving background. However, for clustering efficiency, most of them are applied to the optical flow clustering with a few points information, and the obtained motion targets are not complete.

Some scholars combine the optical flow with other algorithms or models to enhance the precision of moving objects segmentation. The optical flow is regarded as a special geometric model AIC (Akaike information criterion) [37]. The AIC relationship function under different motion states is constructed using the ideal optical flow field and the level of noise magnitude; however, the results exhibit a quite noise. Chen M et al. [38] proposed that the image dynamic texture region does not satisfy the two assumptions that consider the optical flow constant brightness and optical flow spatial consistency; therefore, the optical flow residual, which can detect dynamic texture regions such as smoke, flame, and water surface is generated. The optical flow residual is applied to a Gaussian mixture model (GMM) to extract moving targets [39]. Li et al. [40] proposed a target detection method for a moving background; the matching points are calculated according to Shi-Tomasi corner detection and LK sparse optical flow, and the moving targets are extracted using the three-frame difference method. However, due to the LK sparse optical flow method, the extraction result is low, and the extraction result is unideal under complex background conditions.

Our proposed method can perform the optical flow extraction and clustering analysis of moving targets in a HS dense optical flow vector, aiming to achieve high-precision multi-moving objects extraction and recognition, and is suitable for most optical flow estimation methods. To identify different moving targets regions, this study utilizes K-means clustering to analyse the optical flow clustering of the objects foreground area. In addition, the proposed method combines the influence of optical flow amplitude and direction characteristics on the recognition of different objects.

3 Proposed method

3.1 Enhanced HS optical flow calculation method and moving objects foreground region extraction

The Optical flow method assumes that the brightness of the image is constant, and the spatial position of the same points at different instants of time have changed. This change is represented as the motion vector (also can called as flow direction), which is defined as optical flow.

To enhance anti-interference to the illumination changes and improve robustness rotating deformation in moving objects or the environment area, the proposed optical flow estimation method includes the following two components: data item and smoothing item. The data item includes the image brightness constant assumption and image first-order gradient constant assumption. This design can be insensitive to the change in image brightness, and is suitable for any motion. The optical flow residual term is introduced to the data item to enhance the anti-interference for the noise and dynamic texture. In addition, we utilize the occlusion determination based on Delaunay triangulation in the data item to reduce the influence of the ‘ghost’ problem. For the optical flow smoothing term, we utilize the decreasing scalar function [41] to solve the following problems: the exceeding smoothness of the function, and the fuzziness of the edge. For the moving objects foreground region extraction, we utilize the Harris points in background to achieve the threshold process, which can be utilized in both static and dynamic environments. For the details of the proposed method, we referred to the study conducted by Ding et al. [21]. Although the estimated optical flow field can segment the foreground region of the moving objects accurately with robust point thresholding processing and image morphology processing (the segment results and analysis could be found in Supplement Sect. 1.1). When multiple moving objects in the scene, the segmentation results are the regions of all moving objects, the optical flow information of the image is not fully utilized to achieve objects detection and segmentation.

3.2 Multiple moving objects extraction and recognition based on the HS optical flow histogram and K-means clustering

To realize the recognition of multiple objects scenarios, we further refine and segment the moving objects foreground region according to the different motion information of different objects. One of the most important problems to realize the recognition and segmentation of different moving objects is to find the corresponding feature spatial relations of objects regions. When it can accurately and efficiently identify the number and the member of the different moving objects clusters, we think the spatial relationship is appropriate. The optical flow vectors belonging to different objects are different, whereas the optical flow vectors pertaining to pixels of the same moving objects are similar. Therefore, we use the amplitude feature and direction feature of the image dense optical flow to represent. According to the foreground region optical flow field, similar optical flow clustering is performed to obtain optical flow clusters of different moving targets. Similar optical flow clustering is performed according to the optical flow vectors of each pixel in the foreground region; thus, optical flow clusters are obtained. We calculate the clustering distance in the moving objects foreground regions; the weighted calculation is conducted based on the optical flow amplitude and on the direction of the optical flow vector at different points. To ensure moving objects foreground region segmentation and extraction accuracy, we utilize the K-means method for clustering. The specific process is as follows:

We calculate the optical flow gray value $\mathrm{V}(\mathrm{x}, \mathrm{y})$ using the optical flow $\overrightarrow{\omega}(\mathrm{x}, \mathrm{y}) = [u \;\; v]$. The function is expressed as follows:

$$\mathrm{V}\left(\mathrm{x}, \mathrm{y}\right) = \sqrt{u(x, y)^2 + v(x, y)^2} \tag{1}$$

Subsequently, we normalize $\mathrm{V}(\mathrm{x}, \mathrm{y})$, and the normalized gray image V_{norm} is obtained:

$$V_{norm}\left(\mathrm{x}, \mathrm{y}\right) = \frac{\mathrm{V}\left(\mathrm{x}, \mathrm{y}\right) - V_{min}}{V_{max} - V_{min}} \tag{2}$$

$$\text{s.t. } V_{max} = \max\left(\sqrt{u\left(x, y\right)^2 + v\left(x, y\right)^2}\right)$$

$$V_{min} = \min\left(\sqrt{u\left(x, y\right)^2 + v\left(x, y\right)^2}\right)$$

Subsequently, the threshold of the optical flow is processed; furthermore, the segmentation threshold between the moving objects region and the background region is calculated as per the gray image V_{norm}, and the coarse moving objects detection is obtained.

The optical flow amplitude is the velocity component, which represents the intensity of the optical flow at the point, and the direction of the optical flow vector represents the direction of motion. We define the information of each pixel in the moving objects foreground region as follows:

$$f_i = \left(x_i, y_i, u_i, v_i\right)^T \tag{3}$$

Let $p_i = \left(x_i, y_i\right)^T$ represent the coordinate information, and let $\omega_i = \left(u_i, v_i\right)^T$ represent optical flow vector information. Subsequently, the calculation formula of the optical flow optical flow amplitude R_i and the optical flow direction angle $\angle\theta_i$ is expressed as follows:

$$\begin{cases} {R_i}^2 = {u_i}^2 + {v_i}^2 \\ \angle{\theta_i}^2 = (arc\tan\frac{v_i}{u_i})^2 \end{cases} \tag{4}$$

Optical flow clustering includes the following two steps: clustering centre selection and iteratively updating clustering. The details are as follows:

3.2.1 Cluster centre selection

Because the initial cluster centres selection and the number of cluster centres in K-means clustering exert an immense influence, we generate histograms of optical flow amplitude and the optical flow direction in the objects foreground region. The set corresponding to the optical flow amplitude histogram is expressed as $F_R = (f_{R,1}, f_{R,2}, f_{R,3}, \cdots, f_{R,M})$; the set corresponding to the optical flow direction histogram is expressed as $F_\theta = (f_{\theta,1}, f_{\theta,2}, f_{\theta,3}, \cdots, f_{\theta,N})$; and the initial centre of the cluster is the set of centres selected from F_R and F_θ is expressed as $C^0 = (C_1^0, C_2^0, \cdots C_k^0)$, where k denotes the number of cluster centres.

Based on the Euclidean distance, the distance between each pixel in the foreground area of the moving objects and each cluster centre is calculated as $D_p^2 = (x_i - x_{c_k^0})^2 + (y_i - y_{c_k^0})^2$. According to the shortest distance principle, a cluster set is formed, and k clusters are obtained as $\mathrm{X} = (X_1^0, X_1^0, \cdots X_k^0)$. Subsequently, all clusters X_k^0 are determined. When the number of points in the cluster is exceedingly small, the corresponding cluster centre C_k^0 is deleted. Repeat the preceding process, and update the cluster centre until the cluster centre do not change or the maximum number of iterations is attained (The maximum number of iterations in this paper is 100), and utilize it as the cluster centre for optical flow clustering.

3.2.2 Iterative update clustering

The distance between each cluster points and the cluster centre is calculated by weighting, and the nearest weighted cluster centre is chosen as the principle; thus, the cluster set is formed. The calculation formula of the weighted distance is expressed as follows:

$${D_w}^2 = w_R {D_R}^2 + w_\theta {D_\theta}^2 \tag{5}$$

where D_R denotes the distance of optical flow amplitude between two points, D_θ denotes the angle of optical flow between two points, and w_R and w_θ denote their weight, respectively. Moreover, the calculate function of ${D_R}^2$ and ${D_\theta}^2$ is expressed as follows:

$$\begin{cases} {D_R}^2 = (u_i - u_j)^2 + (v_i - v_j)^2 \\ {D_\theta}^2 = (arctan\frac{v_i}{u_i} - arctan\frac{v_j}{u_j})^2 \end{cases} \tag{6}$$

The weight ratio of the amplitude item and the angle item is defined as 1:1. However, due to the different calculation units of D_R and D_θ, the optical flow amplitude item and the angle item should be normalized. The normalization process entails making the expected value of $w_R {D_R}^2$ and $w_\theta {D_\theta}^2$ the same by confirming the weight values w_R and w_θ, which must meet the following requirements:

$$\mathrm{E}\left(w_R {D_R}^2\right) = \mathrm{E}\left(w_\theta {D_\theta}^2\right) \tag{7}$$

If the normalization of the ${D_R}^2$ item is considered, the expectations is expressed as follows: $\mathrm{E}\left({D_R}^2\right) = E\left((u_i - u_j)^2 + (v_i - v_j)^2\right) = 4E\left({u_i}^2\right) = 4$. Subsequently, if ${D_\theta}^2$ is considered, the angle between two optical flow vectors is $\theta \in (0, \pi)$ and the expectation is expressed as follows: $\mathrm{E}\left({D_\theta}^2\right) = \pi^2/3$, define $w_R = 1$, because $w_R\mathrm{E}\left({D_R}^2\right) = w_\theta \mathrm{E}\left({D_\theta}^2\right)$, we can calculate $w_\theta = 12/\pi^2$. Therefore, the w_R and w_θ in Eq. (5) are expressed as follows: $w_R = 1$, $w_\theta = \frac{12}{\pi^2}$.

After the cluster sets are obtained, the average value of each cluster is selected as the new cluster centre to cluster again. When the cluster centre no longer changes or when the maximum number of iterations is attained (we set the maximum number of iterations to 100), the iteration is stopped, and the optical flow clustering segmentation result of the moving objects is obtained. The clustering results of multiple moving objects and analysis could be found in Supplement Sect. 1.2.

4 Experiment and analysis

4.1 Multiple moving objects detection experimental context and evaluation criterion

4.1.1 Experimental context

We apply the image optical flow estimation of commonly utilized public data sets, such as the MPI Sintel Flow Dataset data set (http://sintel.is.tue.mpg.de/) and KITTI Flow 2015 data set (http://www.cvlibs.net/datasets/kitti/); thus, we verify the effectiveness of the proposed method. The MPI Sintel Flow Dataset is a multi-frame sequential RGB image with a 326×1024 resolution, and the KITTI Flow 2015 dataset is a two-frame sequential RGB image with a 375×1248 resolution. To verify the applicability and precision of objects recognition to common optical flow estimation methods, we applied the commonly utilized LK optical flow [42], HS pyramid optical flow [43], Black & Anandan optical flow (B&A) [44], and Brox optical flow method [45] to conduct experiments, whereas the prevalent Otsu threshold method was utilized for comparison.

4.1.2 Evaluation criterion

The performance test of moving objects recognition and segmentation includes three indexes: E_{Pre} (Precision), E_{Rec} E (Recall), and E_{FM} (F-measure). The calculation formula is expressed as follows:

$$E_{Pre} = \frac{TP}{TP + FP} \tag{8}$$

$$E_{Rec} = \frac{TP}{TP + FN} \tag{9}$$

$$E_{FM} = \frac{2 * E_{pre} * E_{rec}}{E_{pre} + E_{rec}} \tag{10}$$

TP (True Positive) denotes the number of correctly extracted pixels of moving objects; FP (False Positive) denotes the number of extraction error pixels of background environment as the foreground; and FN (False Negative) denotes the number of extraction error pixels of foreground moving objects as the background.

4.2 Experiments analysis

The LK, HS, B&A, Brox and our optical flow estimation compaison and accuracy evaluation could be found in supplement Sect. 2.2.1. Based on this, we compared the Otsu thresholding method in multi-targets detection. Figure 1 illustrates the results of optical flow thresholding segmentation for robust point optical flow estimation. The first line indicates the foreground area of the current frame image and the corresponding real moving objects, whereas the second to fifth lines indicate the LK, HS pyramid, B&A, Brox, and the thresholding processing results of the optical flow method. Column (I) represents the extraction results of Otsu optical flow thresholding, and column (II) depicts the moving objects extraction results of the robust points thresholding processing method. According to the experimental comparison results, KITTI_1 and KITTI_2 sequences are rigid moving objects, and the LK optical flow method is not applicable. In KITT_1, HS pyramid, Brox, and Otsu thresholding segmentation results all misdetect part of the background region as the foreground region, and the detection

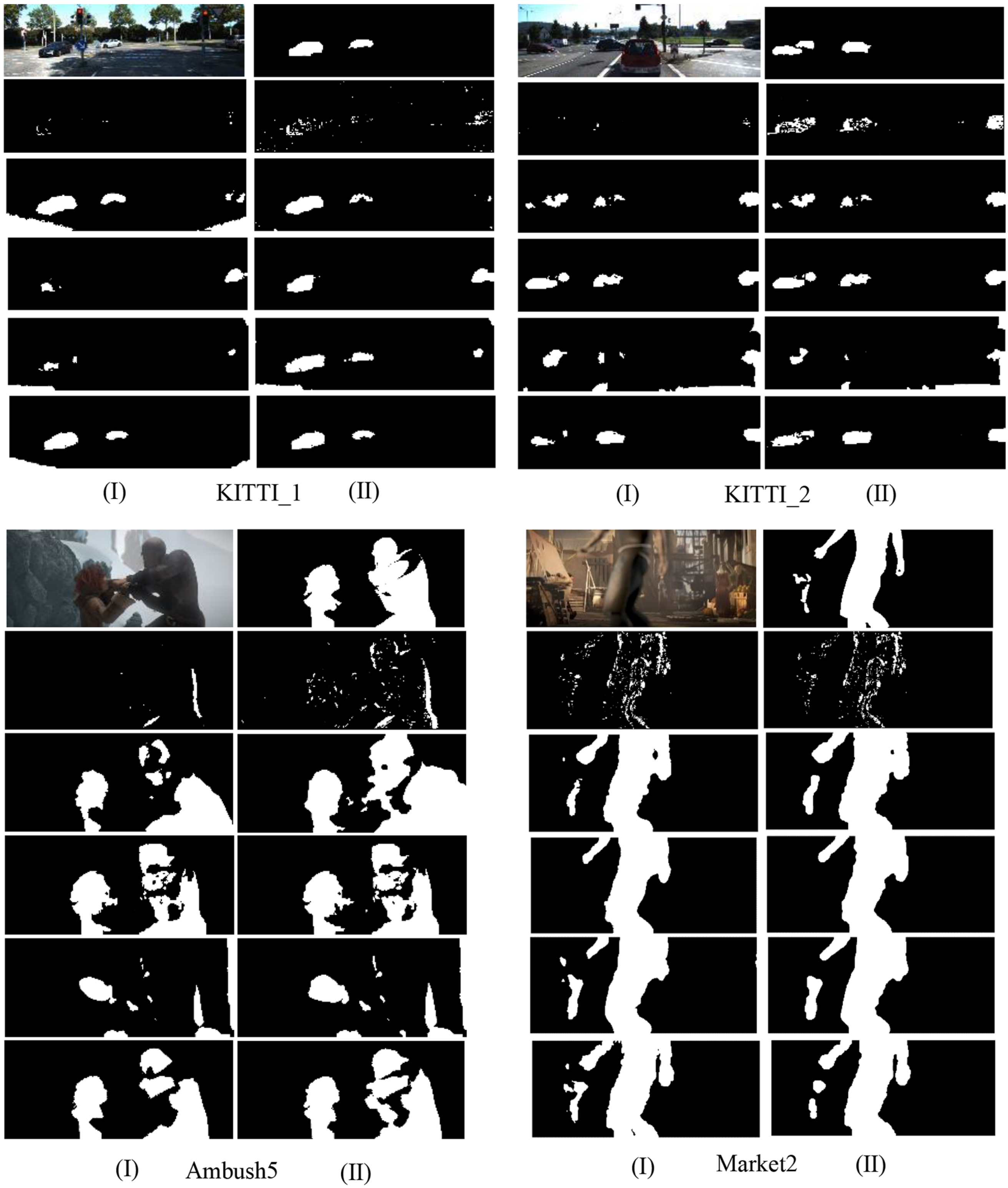

Fig. 1 The thresholding process results are comparable to those of Otsu with HS, B & A, Brox, and the proposed method and their morphological operation results. (**I**) Otsu results. (**II**) the thresholding results of the proposed method. From top to bottom, we can observe the LK result, HS result, B&A result, Brox result, and the result of the proposed method

area is too large to be removed by subsequent morphological operations, whereas the moving target region extracted by the B&A optical flow method using the two segmentation methods is much smaller than the real moving objects regions. In KITTI_2, there is little difference between the detection results of Otsu and that of the proposed method due to the absence of light flow in the background region. In Ambush5 and Market2, due to the large difference in the amplitude and direction of optical flow in the foreground area, detecting the absolutely complete objects foreground area became harder. However, the real foreground region herein relies on the real image optical flow value to extract the foreground region. Therefore, the optical flow amplitude in the foreground area is less or equal to the part in the background region.

Figure 2 depicts the performance comparison of Precision, Recall, and F-measure. As the results show, in KITTI_1 data set, the Precision of OTSU processing by HS, brox, and our optical flow estimation method are low, because of its high false detection rate. In B&A and brox Otsu processing, Recall and F-measure are both low, which is due to the high rate of missed detection. In KITTI_2 and Ambush_5 data sets, the Brox method has low evaluation values in all performance, which is caused by too much missed and false detection cases. These three evaluation indicators can to some extent represent the detection performance of moving objects. But in KITTI_2, Market2 data sets. Some methods have obvious omissions mistakes and still have good evaluations.

Figure 3 depicts the comparison of the results after image morphology operation and the mask display of the proposed multiple moving objects refining segmentation method. From top to bottom, we can observe the current image and the multiple moving objects mask; the HS pyramid; B&A; Brox; and the enhanced optical flow method after Otsu thresholding (left), after robust point threshold segmentation, and after optical flow clustering (right). It can be observed that the proposed multi-objective thinning segmentation based on the optical flow histogram and K-means clustering method exhibits a more optimal clustering effect on rigid moving objects (KITTI_1 and KITTI_2 sequence images). However, the method exhibits errors in clustering

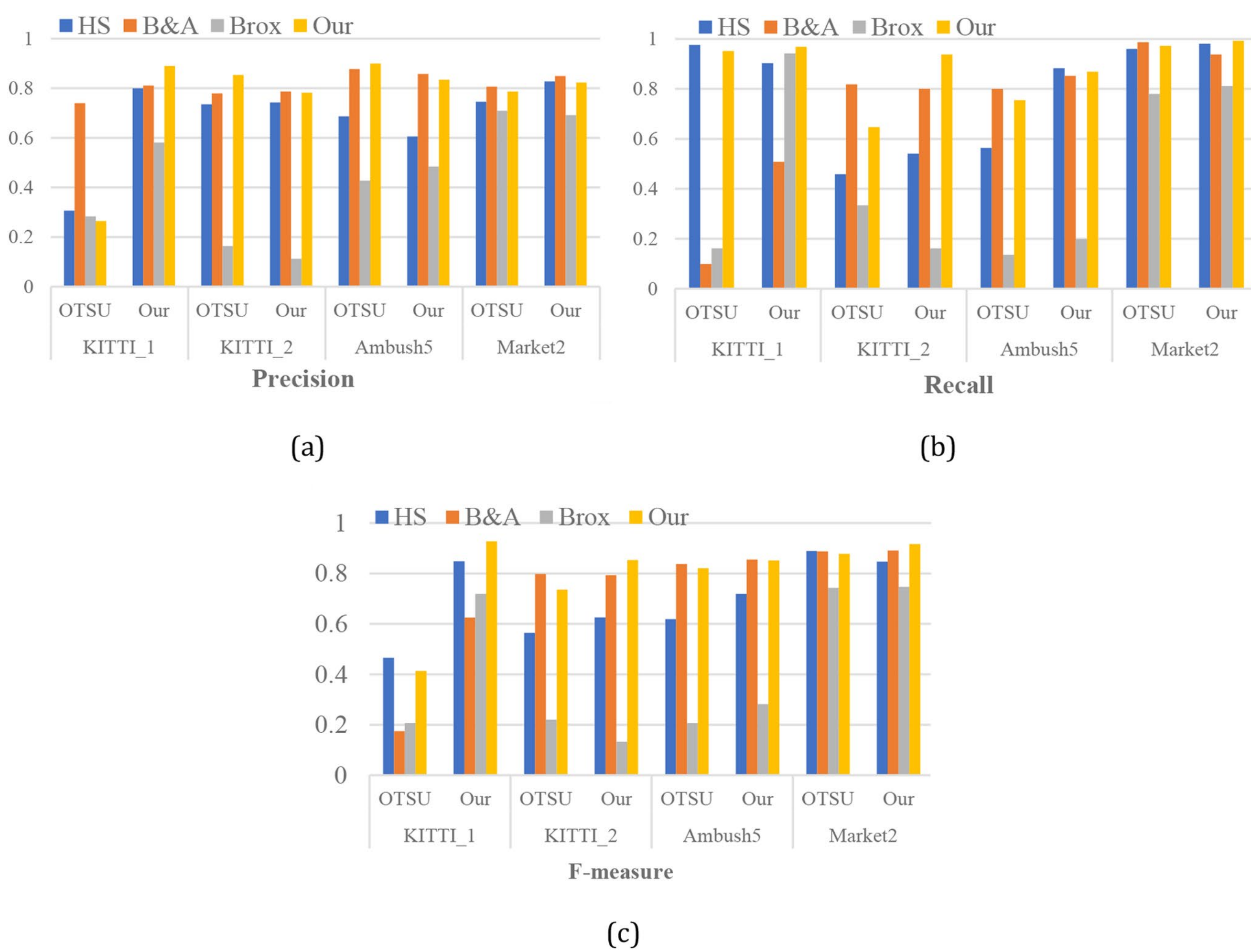

Fig. 2 Comparison of evaluation criterion for extracting the multiple moving objects index

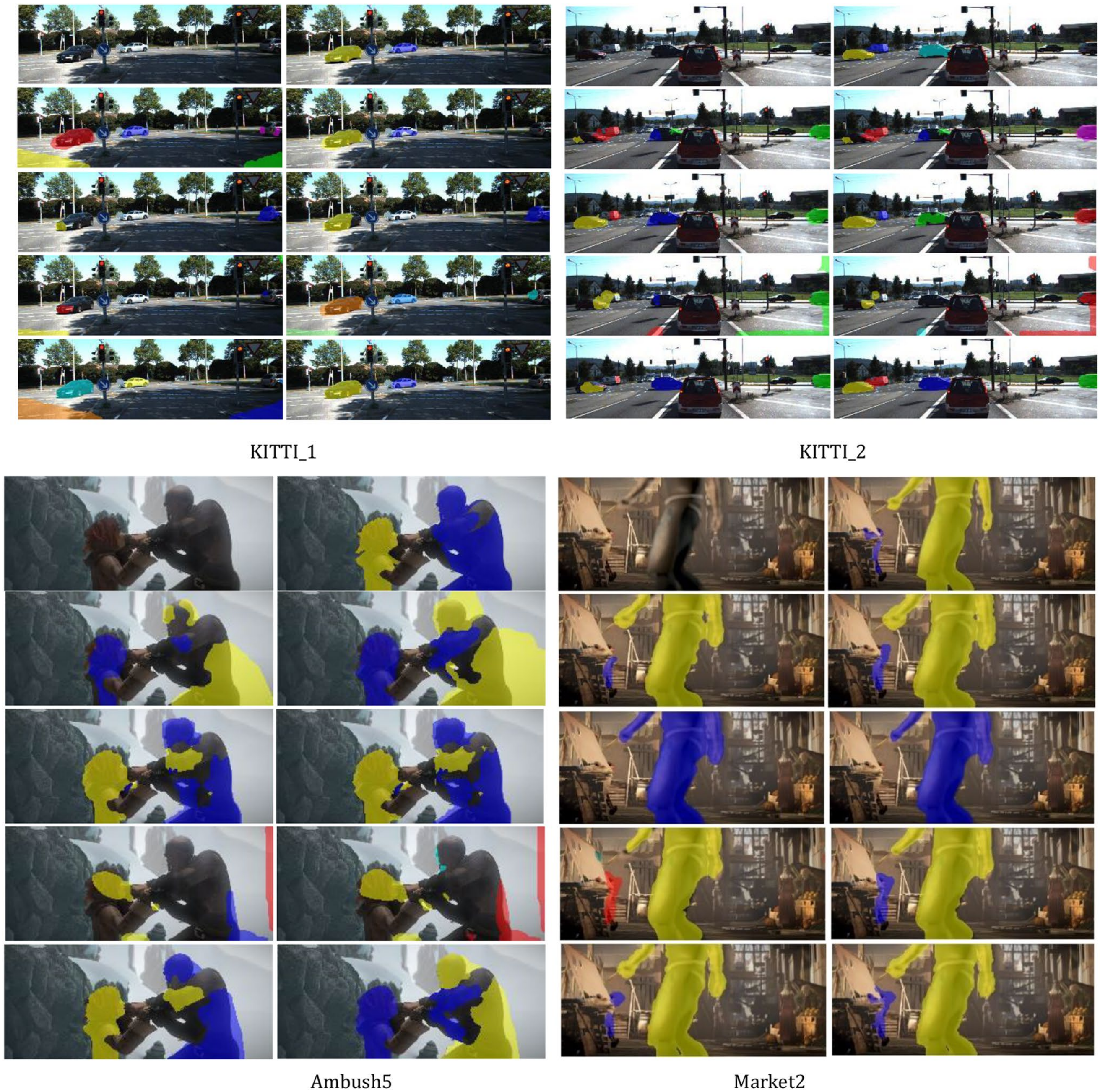

Fig. 3 The mask of multiple moving objects segmentation using optical flow clustering in foreground regions

non-rigid moving objects with large motion difference in the same object (Ambush5). This observation may be rationalized as follows: the optical flow clustering algorithm is based on the similarity between the amplitude and size of the optical flow inside the moving objects area. The same rigid moving object internal optical flow vectors are similar. However, Ambush5 is a non-rigid moving target, and the optical flow at the arm of the moving objects on the right

Table 1 The proportion of correctly detecting multiple moving objects

	KITTI_1		KITTI_2		Ambush5		Market2	
	OTSU	Our	OTSU	Our	OTSU	Our	OTSU	Our
HS	40%	100%	20%	20%	50%	50%	100%	100%
B&A	50%	50%	75%	75%	50%	50%	50%	50%
Brox	33.33%	50%	25%	25%	0%	0%	50%	100%
Our	50%	100%	75%	75%	50%	50%	100%	100%

side is more similar to the left side moving target; therefore, an error classification area exists in the clustering result. The extraction results of targets with similar optical flow inside the same target are depicted in Supplement Fig. 1. For non-rigid moving objects with similar optical flow, the clustering results are more accurate.

Because using only Precision, Recall and F-measure evaluation indexs cannot fully represent the accuracy, when the accuracy of a moving object detection and segmentation reaches 70%, it is considered that the moving object is correctly detected, and based on this, the proportion of the algorithm that can correctly detect moving objects is determined. The results are shown in Table 1.

5 Conclusion

Herein, multiple moving objects recognition and segmentation based on an optical flow histogram and K-means clustering algorithm is analysed. The proposed method exhibits a more optimal recognition and segmentation result for adjacent different moving objects segmentation, which also solves the following problem: the traditional optical flow thresholding method cannot further identify and segment different moving objects. The proposed algorithm exhibits a satisfactory segmentation effect on rigid moving objects and on non-rigid moving targets with small internal optical flow differences in static or moving backgrounds; however, for non-rigid moving objects with large internal optical flow differences in the same object range, the results are not satisfactory.

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Data availability Publicly available datasets were analyzed in this study. This data can be found here: MPI Sintel Flow Dataset data set (http://sintel.is.tue.mpg.de/) and KITTI Flow 2015 data set (http://www.cvlibs.net/datasets/kitti/) (accessed on 10 September 2023).

Declarations

Ethical approval "Not applicable" for studies not involving humans or animals.

Competing interests The authors declare no competing interests.

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